

CAMPUS TWIN

Unified Digital College ERP System — Complete Features Summary Report

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1. CORE SECURITY & AUTHENTICATION MODULES		
1.	Secure Appwrite Authentication Unified login portal using secure Appwrite session creation for students, teachers, mentors, and administrators.	Major
2.	First-Time Login Force Password Change Forces newly created users to update their password immediately upon their first login to ensure account security.	Major
3.	Forgot Password OTP Verification Flow A multi-step recovery flow allowing users to reset their forgotten passwords using a secure 6-digit email verification code.	Major
4.	Secure Account Deletion Serverless API A Netlify serverless function that deletes user profiles and database records securely without exposing credentials.	Major
5.	Multi-Role Route Authorization (Private Routes) Strict UI-level and route-level authorization restricting pages based on user roles (Student, Teacher, Mentor, Admin).	Major
6.	Session Persistence Manager Automatic theme and login session conservation integrated with browser local storage.	Simple
7.	Interactive OTP Demo Mode Simulation Falls back to displaying the generated OTP code directly on the UI when SMTP services are unconfigured, enabling seamless developer testing.	Simple
8.	Global Toast Alerts Integration Integrated React Hot Toast for displaying beautiful, non-intrusive status notifications and warning messages.	Simple
9.	Role-Based Redirect Engine Validates login profiles and automatically redirects users to their respective home dashboards upon entering the domain.	Simple
2. STUDENT PORTAL FEATURES		
10.	Interactive Student Home Dashboard A grid-based dashboard presenting current day lectures, quick statistics, announcements, and navigation widgets.	Major
11.	Interactive Timetable Grid Viewer Displays class timetables with customizable colors and clean layout representations.	Major
12.	Semester Course Registration System Allows students to register for course packages and electives based on branch and semester restrictions.	Major

13.	AICTE Activity Points Tracker Comprehensive panel for students to log AICTE activity points and upload supporting certificates.	Major
14.	Subject-Wise Internal Marks Sheet Displays marks scored in internal assessments, test series, and practical evaluations.	Minor
15.	Academic Marks Card PDF Exporter Generates and exports official university marks cards in PDF format directly from the browser.	Minor
16.	Interactive Calendar of Events A visual timeline presenting academic cycles, holidays, events, and examination dates.	Minor
17.	Attendance Percentage Progress Bar Displays overall and subject-wise attendance with warning indicators if it falls below 75%.	Minor
18.	Real-Time Chat Notification Feed Displays a count of unread messages and updates from the class group chat rooms.	Simple
19.	Direct Portal Profile View Displays student roll number, USN, division, and official department records.	Simple
20.	Class Announcement Bulletin Board A clean notification panel highlighting warnings, notices, and test schedules.	Simple
21.	Live Session Highlight Detects the current time and highlights the ongoing lecture slot in the student dashboard.	Simple
22.	PWA Theme Switcher Integration Allows students to toggle between clean Light Mode and elegant Dark Mode themes.	Simple
23.	LinkedIn Official Profile Redirect Sidebar quick-link redirecting students to their college official LinkedIn pages.	Simple
24.	Official College Website External Redirect Sidebar quick-link redirecting students to the main institutional website.	Simple

3. COMPLAINT BOX & ANTI-RAGGING HOTLINE

25.	Anonymous Student Complaint Box Allows students to post grievance issues, submit feedback, or raise complaints directly to their mentors/admins.	Major
26.	Complaint Ticket Status Tracker A visual timeline representing whether a complaint is Open, In-Progress, or Resolved.	Minor
27.	Admin Complaint Resolver Interface Allows administrators to filter complaints by status, assign resolvers, and post closing feedback.	Minor

28.	Official Anti-Ragging Website Link Integrated national anti-ragging portal redirect link under the complaint box for safety.	Simple
29.	Institutional Drug Prevention Portal Link Direct access link to government drug abuse prevention databases and local helplines.	Simple

4. TEACHER & FACULTY PORTAL FEATURES

30.	Teacher Personal Dashboard A dedicated layout showing assigned subjects, ongoing schedules, and quick portal links.	Major
31.	Class Attendance Logging Interface A checklist grid allowing teachers to select a branch, class, and mark students present or absent.	Major
32.	Student Marks Upload Board Enables teachers to select tests, input grades, and publish scores to the student marks dashboard.	Major
33.	Personal Leave Request Form Allows teachers to apply for casual, medical, or official leaves directly to branch heads.	Minor
34.	Leave Status Tracking History Displays previous leave requests, approval notes, and statuses (Approved/Pending/Rejected).	Minor
35.	Teacher Schedule Planner An agenda view mapping out a teacher's weekly lecture hours and lab allocations.	Minor
36.	Interactive Classroom Chat Room Access Allows faculty members to moderate and post in class chat rooms for their subjects.	Minor
37.	Department Complaint Log Access Enables teachers to view complaints raised by students assigned to their courses.	Simple

5. MENTOR & ADVISOR PORTAL FEATURES

38.	Mentor Group Dashboard A dedicated workspace listing all mentees assigned to the faculty member.	Major
39.	Student Academic Profile Lookup A search system to inspect a mentee's full academic record, including GPA and marks cards.	Major
40.	AICTE Points Verification Queue Allows mentors to view certificates uploaded by their mentees and approve/reject points.	Major
41.	Mentee Attendance Auditor An analysis screen showing detailed student attendance summaries across all semesters.	Minor

42.	Unified Mentoring Record Book Allows advisors to log notes, review semester registrations, and maintain student histories.	Minor
43.	Exam History Audit Log Allows advisors to view previous backlog details, SGPA history, and student CGPA progress.	Minor
44.	Direct Mentor-Mentee Announcement Tool Sends notification feeds specifically to a mentor's assigned group of student users.	Simple

6. ADMIN ERP & CAMPUS CONFIGURATIONS

45.	AI-Assisted Automated Timetable Generator An intelligent scheduling engine that automatically calculates section timetables while avoiding clashes.	Major
46.	Curriculum & Subject Database Manager Enables admins to define course codes, names, credits, and lab integrated status.	Major
47.	User Account Creation Dashboard Enables adding student, teacher, or mentor logins, assigning their branch, division, and USN.	Major
48.	Branch Configuration Setup Panel Enables managing branches (e.g. CSE, ECE) and editing institutional metadata.	Major
49.	Class Configuration Setup Panel Enables configuring sections, semesters, advisors, mentors, and credit limits.	Major
50.	Dynamic Time Slot Manager Allows branch admins to customize timetable period start/end timings per class.	Minor
51.	Excel Bulk Student Importer (.xlsx) Allows importing student rosters in bulk using pre-formatted Excel sheets.	Minor
52.	Excel Blank Template Exporter Downloads clean Excel template sheets for bulk student roster preparation.	Minor
53.	Double-Click Timetable Preview Overrides Allows overriding generated timetable options by double-clicking cells before database save.	Minor
54.	Zero-Credit Subject Configuration Fully supports zero-credit (audit/mandatory non-credit) subjects in curriculum setup.	Minor
55.	Copy Class ID Clipboard tool A quick-action button in Class Management to copy Appwrite Class IDs instantly.	Simple
56.	Event & Announcement Publisher Allows posting college-wide alerts, test sheets, or event listings to the bulletin.	Simple

7. HOSTEL MANAGEMENT SYSTEM (ERP)

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| 57. | Hostel Selection Gateway | Major |
| Allows logging into either Boys' or Girls' hostel divisions based on student roles. | | |

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| 58. | Warden Administration Portal | Major |
| A full dashboard for hostel wardens to allocate rooms, view bills, and approve leave outpasses. | | |

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| 59. | Warden Room Allocation Board | Major |
| A visual manager for hostel room occupancy, detailing room statuses and resident USNs. | | |

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| 60. | Student Hostel Outpass & Leave Form | Major |
| Enables students to submit leave applications (outpasses) specifying destination and dates. | | |

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| 61. | Warden Leave Request Approval Queue | Major |
| Allows wardens to inspect student outpasses and approve or reject them with feedback. | | |

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| 62. | Automated Monthly Bill Generator | Minor |
| Runs a monthly batch calculation to generate mess, rent, and utility bills for residents. | | |

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| 63. | Resident Fee Tracker & Payment status | Minor |
| Displays resident payment history, showing paid, pending, or overdue hostel fees. | | |

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| 64. | Hostel Grievance Board | Minor |
| Allows residents to lodge complaints regarding rooms, Wi-Fi, electricity, or mess quality. | | |

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| 65. | Warden-Student Hostel Chat Rooms | Minor |
| Real-time communication channels for wardens to post alerts and coordinate with residents. | | |

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| 66. | Hostel Notice Board & Regulations Viewer | Simple |
| Publishes emergency warden alerts, rules, and timings directly to the student hostel feed. | | |

8. PLACEMENT & CAREER PORTAL

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| 67. | Placement Selection Gateway | Major |
| Enables logging in as either student applicant or Placement Officer administrator. | | |

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| 68. | Placement Officer Portal | Major |
| A control dashboard for managing campus drives, applicant lists, and job profiles. | | |

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| 69. | Job Posting Board & Drive Creator | Major |
| Enables placement officers to post company details, salary CTC, roles, and eligibility. | | |

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| 70. | Interactive Student Resume Builder | Major |
| An interactive editor for students to input skills, projects, and education to generate a CV. | | |

71.	Student Job Application Center Displays active campus drives matching eligibility criteria with a one-click apply button.	Major
72.	Resume PDF Exporter Generates a clean PDF formatting of the student's resume for campus placement purposes.	Minor
73.	Applicant Shortlist Tracker Displays students who applied for drives and allows shortlisting for interviews.	Minor
74.	Interview Calendar & Schedule Notifications Alerts candidates about drive rounds, written tests, and technical interview timings.	Minor
75.	Drive Candidate Excel Exporter Enables placement officers to export shortlisted students to an Excel sheet for HR teams.	Minor
76.	Placement Drive History A directory tracking previous companies visited, offers made, and student selections.	Simple

9. COPILOT AI ASSISTANT & CHATBOT WIDGET

77.	Copilot AI Chatbot Widget A chatbot floating widget available across dashboards to assist with campus questions.	Major
78.	AI Text-to-Speech Voice Synthesis Allows the AI chatbot to read responses aloud using system voice synthesizers.	Minor
79.	AI Response Copy to Clipboard One-click button on each chatbot reply to instantly copy text responses.	Simple
80.	Floating Quick Assistant Panel A floating button that opens a beautiful modal chat overlay with one click.	Simple

10. CHAT, PWA & ERP UTILITIES

81.	Real-Time Class Chat Rooms Real-time, WebSocket-based group messaging boards for every registered classroom section.	Major
82.	Official Documents Share & Approvals Allows students to share official docs (ODs, certificates) for approval by mentors.	Major
83.	Anti-Screenshot Secure Document Shield Protects sensitive files inside the Document Cabinet from screenshots using instant window blur filters, print blockers, and disabled context menus.	Major
84.	ERP System Maintenance Mode Screen Blocks non-administrative access and displays a clean status screen during ERP updates.	Minor

85. Initials-Based Avatar Fallback Generator

Simple

Automatically renders initials placeholders for user profiles without uploaded photos.

86. PWA Service Worker Offline Support

Simple

Registers a service worker to cache pages, stylesheet assets, and index structures.

87. Responsive Sidebar Layout

Simple

A collapse-ready dashboard navigation sidebar optimized for mobile, tablet, and desktop.

88. Timetable Touch Scroll Integration

Simple

Allows responsive horizontal touch-swipe scrolling on mobile to view timetable columns.
